

## Demonstration of Proposed Features

|                      |                                       |
|----------------------|---------------------------------------|
| <b>Date</b>          | 11 July 2026                          |
| <b>Team ID</b>       |                                       |
| <b>Project Name</b>  | A Comprehensive Measure of Well-Being |
| <b>Maximum Marks</b> | 1 Mark                                |

### Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

| S.No | Feature Name          | Description                                                                              | Status (Implemented / Partial / Pending) | Demonstrated (Yes / No) | Remarks                                                                 |
|------|-----------------------|------------------------------------------------------------------------------------------|------------------------------------------|-------------------------|-------------------------------------------------------------------------|
| 1    | User Input Form       | HTML/CSS interface allowing users to input the 4 core socio-economic metrics.            | Implemented                              | Yes                     | Fully functional and styled on the live website.                        |
| 2    | Input Validation      | Logic to ensure entered values are within logical bounds before processing.              | Implemented                              | Yes                     | Prevents backend crashes from invalid data entries.                     |
| 3    | Data Normalization    | Processes and scales the raw user inputs to match the ML training data format.           | Implemented                              | Yes                     | Operates invisibly in the Flask backend using <code>scaler.pkl</code> . |
| 4    | HDI Prediction Engine | Executes the Scikit-learn Linear Regression model to calculate the final HDI score.      | Implemented                              | Yes                     | Successfully predicts scores with high accuracy ( $R^2 = 0.981$ ).      |
| 5    | Results Dashboard     | Dynamically displays the predicted HDI score and development tier category.              | Implemented                              | Yes                     | Updates seamlessly via JavaScript without reloading the page.           |
| 6    | REST API Endpoint     | Flask routing ( <code>/predict</code> ) that receives JSON data and returns predictions. | Implemented                              | Yes                     | Verified via Postman and frontend integration.                          |

**Feature Implementation Summary:**

| Metric                          | Count / Value |
|---------------------------------|---------------|
| Total Features Proposed         | 8             |
| Total Features Implemented      | 7             |
| Total Features Demonstrated     | 7             |
| Overall Implementation Rate (%) | 87.5%         |